INSTALLATION GUIDE

HydraBot Stack

Multi-agent AI infrastructure powered by Claude Code CLI

Version **1.0** Platform **Ubuntu 22.04 / Debian 12** Requires **Claude Max subscription**

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0 Overview

HydraBot is a deployable AI agent stack. Each bot is a Telegram interface powered by Claude Code CLI, connected to a shared memory system (AI-IQ) and a multi-agent mesh (Circus). Ephemeral sub-workers handle background tasks in parallel.

```
Telegram API │ ▼ PM2 Bots (grammy + Claude Code CLI) │ ├── circus-bridge.mjs ──►
Circus API :6200 (FastAPI, systemd) │ │ │ ▼ │ AI-IQ memory (SQLite + vectors) │ └──
bot-circus/dispatch.mjs ──► Ephemeral Claude workers (up to 10 parallel)
```

LAYER	COMPONENT	PURPOSE
0	`Claude Code CLI`	AI inference engine — all bots call this
1	`AI-IQ`	Per-agent long-term memory
2	`Circus`	Agent mesh: registry, rooms, task queue
3	`Bot-Circus`	Ephemeral worker pool for sub-tasks
4	`circus-bridge.mjs`	Node.js ↔ Circus integration module
5	`Bots`	Telegram interface + business logic

1 Prerequisites

REQUIREMENT	MINIMUM	NOTES
OS	Ubuntu 22.04 / Debian 12	macOS with Homebrew also works
RAM	1 GB	2 GB recommended if running multiple bots
Disk	5 GB free	10 GB recommended
CPU	1 core	2 cores recommended
Node.js	22.x	Installer handles this automatically
Python	3.10+	Installer handles this automatically
Claude subscription	Claude Max	**Required.** claude.ai/upgrade — used for auth
Telegram bot token	One per bot	Free — create via @BotFather

⚠️ **Claude Max is required**

The Claude Code CLI authenticates against your claude.ai account. A Max subscription is needed for the API access the bots rely on. The installer will pause and prompt you to log in if not already authenticated.

2 Clone the Repository

```
# Clone to /opt/hydrabot (recommended) or any directory
git clone <repo-url> /opt/hydrabot
cd /opt/hydrabot
```

ℹ️ You can install to any path. The installer respects the `HYDRABOT_DIR` environment variable. Examples in this guide use `/opt/hydrabot`.

3 Run the Installer

```
# Run as root (or with sudo)
sudo HYDRABOT_DIR=/opt/hydrabot bash /opt/hydrabot/deploy/install.sh
```

The installer performs these steps automatically:

1. Installs system packages: `curl git python3 python3-pip sqlite3 build-essential`
2. Installs Node.js 22 via NodeSource (skips if already installed)

3. Installs PM2 globally and configures systemd startup
4. Installs Claude Code CLI via npm (skips if already installed)
5. Installs Python packages: `ai-iq circus-agent`
6. Bootstraps the Circus data directory and generates a keypair
7. Installs bot-circus dependencies and initialises performer workspaces
8. Installs and starts the Circus API as a systemd service

> ⚠️ **Auth pause**
>
> If Claude Code CLI is not already authenticated, the installer will stop and print:
>
> ```
> Claude Code CLI installed but needs authentication.
> Run 'claude' interactively to authenticate, then re-run this script.
> ```
>
> Open a new terminal, run `claude`, complete the browser login with your Claude Max account, then re-run the installer command.

VERIFY INSTALLATION

```
# Circus API should be healthy
curl http://localhost:6200/health
# → {"status": "ok"}

# Claude CLI should respond
claude -p "say hello"

# PM2 should be running
pm2 list
```

4 Configure Your First Bot

GET A TELEGRAM BOT TOKEN

1. Open Telegram and search for **@BotFather**
2. Send `/newbot` and follow the prompts
3. Copy the token — it looks like `7123456789:AAExxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx`

GET YOUR TELEGRAM USER ID

1. Search for **@userinfobot** in Telegram

2 Send it any message — it replies with your numeric ID

CREATE AND CONFIGURE THE BOT

```
# Copy the template
cp -r /opt/hydrabot/bots/template /opt/hydrabot/bots/mybot

# Copy and edit the env file
cp /opt/hydrabot/bots/template/.env.example /opt/hydrabot/bots/mybot/.env
nano /opt/hydrabot/bots/mybot/.env
```

REQUIRED `.ENV` VALUES

VARIABLE	REQUIRED	DESCRIPTION
TELEGRAM_BOT_TOKEN	REQUIRED	Token from @BotFather
ALLOWED_USER_ID	REQUIRED	Your numeric Telegram user ID (from @userinfobot)
BOT_NAME	OPTIONAL	Display name for the bot — default: `MyBot`
CLAUDE_CLI_PATH	OPTIONAL	Path to `claude` binary. Default: `claude` (uses PATH). Get with `which claude`
CLAUDE_WORKING_DIR	OPTIONAL	Working directory for Claude. Default: current directory
CLAUDE_TIMEOUT	OPTIONAL	Response timeout in ms. Default: `120000` (2 min)

START THE BOT

```
# Install dependencies
npm install --prefix /opt/hydrabot/bots/mybot

# Start with PM2
pm2 start /opt/hydrabot/bots/mybot/bot.mjs --name mybot

# Save so it survives reboot
pm2 save

# Watch logs
pm2 logs mybot
```

✅ **Bot is running**

Open Telegram, find your bot, and send it a message. It should respond via Claude within a few seconds.

5 Autonomous Worker (Optional)

The worker polls a job queue and executes long-running tasks via Claude. Useful for background operations triggered from Telegram (`/task` style commands).

```
# Configure
cp /opt/hydrabot/performers/worker/.env.example \
   /opt/hydrabot/performers/worker/.env
nano /opt/hydrabot/performers/worker/.env

# Install deps
npm install --prefix /opt/hydrabot/performers/worker

# Start
pm2 start /opt/hydrabot/performers/worker/worker.mjs --name worker
pm2 save
```

VARIABLE	REQUIRED	DESCRIPTION
TELEGRAM_BOT_TOKEN	OPTIONAL	If set, worker sends job completion messages via Telegram
TELEGRAM_CHAT_ID	OPTIONAL	Your chat ID — where job results are sent
CLAUDE_CLI_PATH	OPTIONAL	Path to `claude` binary
CLAUDE_WORKING_DIR	OPTIONAL	Directory Claude operates in during jobs

6 Running Multiple Bots

Each bot is an independent directory. Repeat the configure step for each one:

```
# Second bot
cp -r /opt/hydrabot/bots/template /opt/hydrabot/bots/designbot
cp /opt/hydrabot/bots/template/.env.example /opt/hydrabot/bots/designbot/.env
# Edit .env with different TELEGRAM_BOT_TOKEN
npm install --prefix /opt/hydrabot/bots/designbot
pm2 start /opt/hydrabot/bots/designbot/bot.mjs --name designbot
pm2 save
```

> ℹ️ All bots share one Claude Code CLI process pool and one Circus mesh. Each bot gets its own Telegram token, allowed user, and performer workspace.

7 Troubleshooting

SYMPTOM	CAUSE	FIX
Bot doesn't respond	Claude not authenticated	Run `claude -p "hello"` manually. If it fails, run `claude login`
`claude: command not found`	CLI not in PATH	Run `which claude` — if empty, run `npm install -g @anthropic-ai/claude-code`. Then set `CLAUDE_CLI_PATH` in `.env`
Circus health check fails	Service not running	`systemctl status circus-api` — if failed, check `journalctl -u circus-api -n 50`
Bot starts then crashes	Missing env var	`pm2 logs mybot` — look for "is required" errors. Check `.env` has `TELEGRAM_BOT_TOKEN` and `ALLOWED_USER_ID`
Auth works locally but not on VPS	SSH session has no browser	Run `claude login` — it prints a URL. Open that URL in any browser to complete OAuth
Circus not connecting	Wrong URL	Default is `http://127.0.0.1:6200`. Set `CIRCUS_URL` in `.env` if Circus runs elsewhere

USEFUL COMMANDS

```
pm2 list                    # all running processes
pm2 logs mybot              # live bot logs
pm2 restart mybot           # restart a bot
pm2 stop mybot              # stop a bot
systemctl status circus-api # Circus API status
curl http://localhost:6200/health # quick Circus health check
claude -p "hello"           # test Claude CLI directly
```

8 Directory Reference

```
hydrabot/
├── deploy/
│   ├── install.sh              # Full VPS installer — run this first
│   ├── circus-api.service      # systemd unit for Circus API
│   └── ecosystem.config.cjs   # PM2 ecosystem config
├── bot-circus/
│   ├── dispatch.mjs            # Ephemeral worker pool (shared by all bots)
│   └── setup-performers.mjs   # Initialise performer workspaces
├── performers/
│   ├── template/               # Blank performer workspace
│   ├── webbs/                  # Frontend designer bot (full example)
│   └── worker/                 # Autonomous job worker
├── bots/
│   └── template/               # ← start here when creating a bot
├── circus-bridge.mjs           # Circus integration (imported by bots)
└── docs/
    ├── architecture.md
    ├── configuration.md
    └── ai-iq.md
```